

One-tail vs. two-tail P values

When comparing two groups, you must distinguish between one- and two-tail P values. Some books refer to one-sided and two-sided P values, which mean the same thing.

What does one-tail mean?

It is easiest to understand the distinction in context. So let's imagine that you are comparing the mean of two groups (with an unpaired t test). Both one- and two-tail P values are based on the same null hypothesis, that two populations really are the same and that an observed discrepancy between sample means is due to chance.

A two-tailed P value answers this question:

Assuming the null hypothesis is true, what is the chance that randomly selected samples would have means as far apart as (or further than) you observed in this experiment with either group having the larger mean?

To interpret a one-tail P value, you must predict which group will have the larger mean before collecting any data. The one-tail P value answers this question:

Assuming the null hypothesis is true, what is the chance that randomly selected samples would have means as far apart as (or further than) observed in this experiment with the specified group having the larger mean?

If the observed difference went in the direction predicted by the experimental hypothesis, the one-tailed P value is half the two-tailed P value (with most, but not quite all, statistical tests).

When is it appropriate to use a one-tail P value?

A one-tailed test is appropriate when previous data, physical limitations, or common sense tells you that the difference, if any, can only go in one direction. You should only choose a one-tail P value when both of the following are true.

- You predicted which group will have the larger mean (or proportion) before you collected any data. If you only made the "prediction" after seeing the data, don't even think about using a one-tail P value.
- If the other group had ended up with the larger mean – even if it is quite a bit larger – you would have attributed that difference to chance and called the difference 'not statistically significant'.

Here is an example in which you might appropriately choose a one-tailed P value: You are testing whether a new antibiotic impairs renal function, as measured by serum creatinine. Many antibiotics poison kidney cells, resulting in reduced glomerular filtration and increased serum creatinine. As far as I know, no antibiotic is known to decrease serum creatinine, and it is hard to imagine a mechanism by which an antibiotic would increase the glomerular filtration rate. Before collecting any data, you can state that there are two possibilities: Either the drug will not change the mean serum creatinine of the population, or it will increase the mean serum creatinine in the population. You consider it impossible that the drug will truly decrease mean serum creatinine of the population and plan to attribute any observed decrease to random sampling. Accordingly, it makes sense to calculate a one-tailed P value. In this example, a two-tailed P value tests the null hypothesis that the drug does not alter the creatinine level; a one-tailed P value tests the null hypothesis that the drug does not increase the creatinine level.

The issue in choosing between one- and two-tailed P values is not whether or not you expect a difference to exist. If you already knew whether or not there was a difference, there is no reason to collect the data. Rather, the issue is whether the direction of a difference (if there is one) can only go one way. You should only use a one-tailed P value when you can state with certainty (and before collecting any data) that in the overall populations there either is no difference or there is a difference in a specified direction. If your data end up showing a difference in the “wrong” direction, you should be willing to attribute that difference to random sampling without even considering the notion that the measured difference might reflect a true difference in the overall populations. If a difference in the “wrong” direction would intrigue you (even a little), you should calculate a two-tailed P value.

How Prism reports one-tail P values

When you ask Prism to report a one-tail P value, it assumes the actual difference or effect went in the direction you predicted, so the one-sided P value reported by Prism is always smaller (almost always, exactly half of) the two-tail P value.

If, in fact, the observed difference or effect goes in the opposite direction to what you predicted, the one-sided P value reported by Prism is wrong. The actual one-tail P value will equal 1.0 minus the reported one. For example, if the reported one-tail P value is 0.04 and the actual difference is in the opposite direction to what you predicted, then the actual one-sided P value is 0.96.

What if you didn't predict the direction of the difference or effect before collecting data?

If you didn't predict the direction of the effect before collecting data, you should not be reporting one-sided P values. It is cheating to say "well, I would have predicted...". If you didn't record the prediction, then you should not use a one-sided P value.

What if there are not two directions to the test?

The concept of one- and two-tail P values only makes sense for hypotheses where there are two directions to the effect, an increase or a decrease. If you are comparing three or more groups (ANOVA), then the concept of one- and two-tail P value makes no sense, and Prism doesn't ask you to make this choice.

How to convert between one- and two-tail P values

If the actual effect went in the direction you predicted:

- The one-tail P value is half the two-tail P value.
- The two-tail P value is twice the one-tail P value (assuming you correctly predicted the direction of the difference).

This rule works perfectly for almost all statistical tests. Some tests (Fisher's test) are not symmetrical, so these rules are only approximate for these tests.

If the actual effect went in the opposite direction to what you predicted:

- The one-tail P value equals 1.0 minus half the two-tail P value.